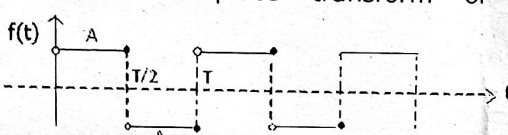




DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CI/CY31	Sub:	Laplace Transforms and Vector space	Test:	01
Semester:	III	Term:	20-11-2023 to 02-03-2024	Marks:	30
Date:	04.01.2024	Time:	03.30 - 04.30 PM		

Note: Answer any TWO full questions. Each main question carries 15 marks

Q.No.	Questions	Blooms Level	CO's	Marks
1. (a)	Write the formula for $L\left(t \int_0^t f(t) dt\right)$	L1	CO1	2
(b)	Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$, then find $L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\}$	L2	CO1	3
(c)	Evaluate i) $\int_0^{\infty} \frac{e^{-3t} \sin t}{t} dt$ (ii) $L\left\{t^9 e^{-\frac{3}{2}t}\right\}$	L3	CO1	5
(d)	Obtain the Laplace transform of the $f(t)$ given by figure 	L5	CO1	5
2. (a)	Define the Dirac-Delta function and sketch its graph.	L1	CO2	2
(b)	Evaluate $L^{-1}\left(s \log\left(\frac{s+4}{s-4}\right)\right)$	L2	CO2	3
(c)	Express the function $f(t) = \begin{cases} t^2, & 0 < t < 2, \\ 4t, & 2 < t < 4, \\ 8, & t > 4. \end{cases}$ in terms of the Heaviside function and hence find its Laplace transform.	L3	CO2	5
(d)	A voltage Ee^{-at} is applied at $t=0$ to a circuit of inductance L and resistance R . Show that the current at time $t=0$ is $\frac{E}{R-al} (e^{-at} - e^{-Rt/L})$ using Laplace transforms.	L4	CO2	5
3. (a)	Define a periodic function with an example.	L1	CO1	2
(b)	Find the inverse Laplace transform of $\frac{1}{s(s^2+1)}$	L2	CO2	3
(c)	If $L\{f(t)\} = F(s)$ then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$, where n is a positive integer.	L3	CO1	5
(d)	Using Laplace transform, solve the simultaneous equation $\frac{dx}{dt} - 2y = \cos 2t$; $\frac{dy}{dt} + 2x = \sin 2t$, given $x(0) = 1$, $y(0) = 0$.	L4	CO2	5